

Emulators for Uncertainty Propagation: From Nuclear Forces to Many-Body Physics

Ryan Curry

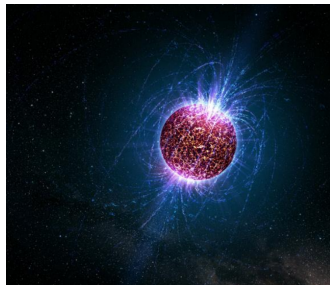
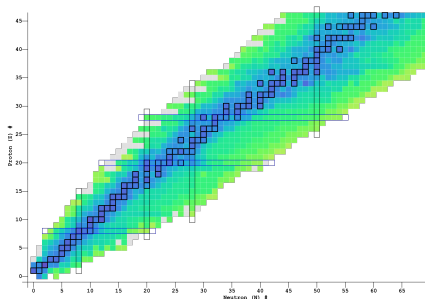
University of Guelph

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2026 Workshop on Progress in Ab Initio Nuclear Theory

The Nuclear Many-Body Problem

From nuclear structure to neutron stars



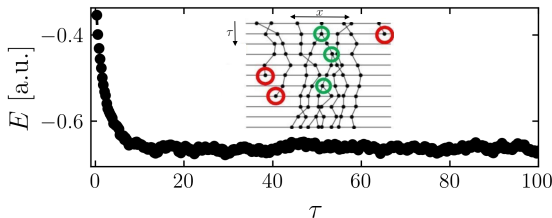
We want to describe diverse nuclear systems from first principles

Diffusion Monte Carlo

Solve Schrödinger equation in imaginary time $-\frac{\partial}{\partial\tau}\Psi = \left(-\frac{\hbar}{2m}\nabla^2 + V\right)\Psi$

$$\psi(\tau) = e^{-(H-E_T)\tau}\psi_T$$

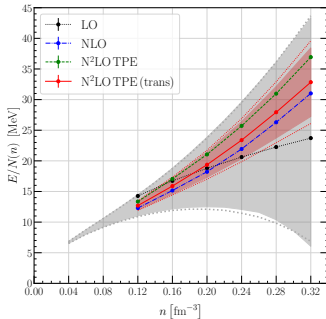
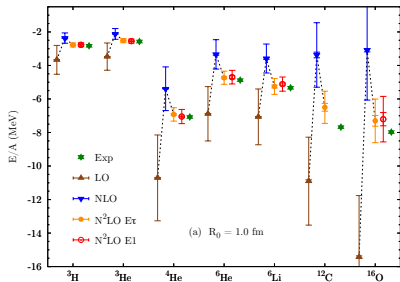
- Projects out the ground-state energy



- Can straightforwardly handle high-cutoff / hard-core interactions
- Works well for central interactions $e^{-\sum_{i<j} v(\mathbf{r}_{ij})\tau}$
- Care needed for spin/isospin dependent interactions (chiral EFT)

Auxiliary Field Diffusion Monte Carlo

- Instead of summing over possible spin-isospin states (GFMC), we sample them with Hubbard-Stratonovich transformations
- Trade off in precision due to sampling auxiliary fields
- Extend calculations to $A \leq 18$ as well as dense matter



D. Lonardoni, S. Gandolfi, J.E. Lynn, *et. al*, Phys. Rev. C. **97**, 044318 (2018)

I. Tews, R. Somasundaram, *et. al*, Phys. Rev. Res. **7**, 033024 (2025)

There are fundamental uncertainties that should be accounted for in a many-body calculation.

- Nuclear Interaction (Experimental / Theoretical)
 - Interactions are fit to experimental measurement (which come with some associated error)
 - Chiral EFT Hamiltonian must be truncated at a finite order
- Many-Body Methods (Computational / Theoretical)
 - Model space truncation (NCSM)
 - Fixed-node / constrained-path approximation (QMC methods)
 - Retaining only two-body operators (IMSRG(2), CCSD)

Uncertainty Propagated Quantum Monte Carlo

- Train emulators on exact few-body calculations
- Bayesian inference fit of all 2N and 3N LECs at N²LO
- Propagate relevant uncertainties to many-body predictions

Quantum Monte Carlo Calculations of Light Nuclei with Fully Propagated Theoretical Uncertainties

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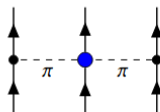
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We report on the first quantum Monte Carlo calculations of helium isotopes with fully propagated theoretical uncertainties from the interaction to the many-body observables. To achieve this, we build emulators for solutions to the Faddeev equations for the binding energy and Gamow-Teller matrix element of ³He, as well as for auxiliary-field diffusion Monte Carlo calculations of the ⁴He charge radius, employing local two- and three-body interactions up to next-to-next-to-leading order in chiral effective field theory. We use these emulators to determine the posterior distributions for all low-energy couplings that appear in the interaction up to this order using Bayesian inference while accounting for theoretical uncertainties. We then build emulators for auxiliary-field diffusion Monte Carlo for helium isotopes and propagate the full posterior distributions to these systems. Our approach serves as a framework for *ab initio* studies of atomic nuclei with consistently treated and correlated theoretical uncertainties.

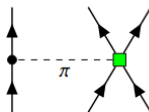
arxiv:2510.015860

Fitting chiral EFT interactions

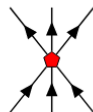
- 9 two-body LECS (C_S, C_T, C_{1-7}) up to N^2 LO fit against experimental phase shift data
- Long-range 3N couplings c_1, c_3 , and c_4 are constrained by π N scattering data.
- Short-range 3N couplings c_D and c_E need to be constrained by few-body observables.



$V_{3N}^{2\pi}(c_1, c_3, c_4)$



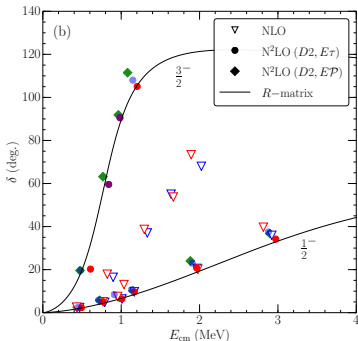
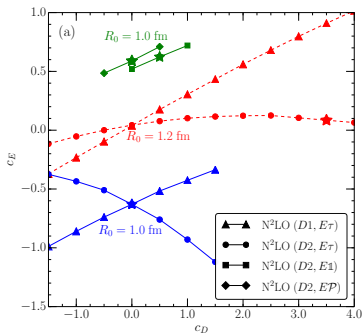
$V_{3N}^{1\pi}(c_D)$



$V_{3N}^{\text{contact}}(c_E)$

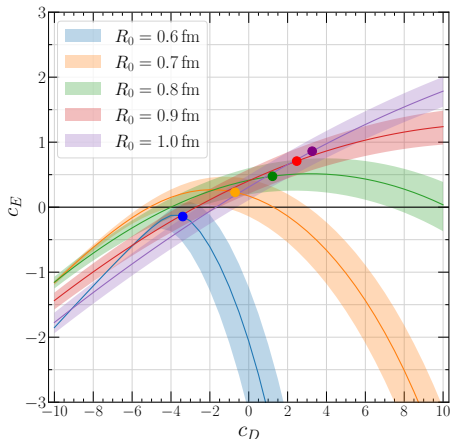
Previously

- c_D and c_E fit to GFMC binding energy of ${}^4\text{He}$ and n - α p -wave phase shifts
- Using a least-squares minimization approach



More recently...

- Fit against Faddeev calculations
- ^3H binding energy and β -decay (similar to Gazit *et. al* 2009)
- These calculations don't mix well with Bayesian inference.
- A single Faddeev calculation takes ~ 1 hour
- Bayesian inference requires 10^5 - 10^6 evaluations



A Tale of Two Emulators

Algorithm that mimics the behavior of a high-fidelity calculation for a fraction of the computational cost.

- In our case:
 - High fidelity means *ab initio* few-body calculation
 - Computational cost $\sim 10^3$ s (Faddeev) or 10^6 s (AFDMC)
- The goal:
 - Eigenvector Continuation (EC)
 - Parametric Matrix Model (PMM)
 - Dramatic reduction in computational cost

D. Frame, R. He, I. Ipsen, D. Lee, D. Lee, and E. Rrapaj Phys. Rev. Lett. **121**, 032501 (2018)

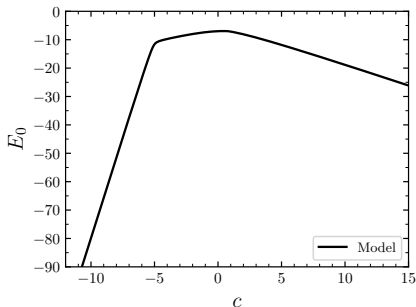
P. Cook, D. Jammooa, M. Hjorth-Jensen, D. Lee, and D. Lee, Nat. Commun. **16**, 5929 (2025)

An example...

Consider a “high-fidelity model” with a single control parameter c

$$H = H^0 + cH^1 \quad (1)$$

We can solve for the lowest eigenvalue of H for any value of c
(but let's pretend it's very expensive)

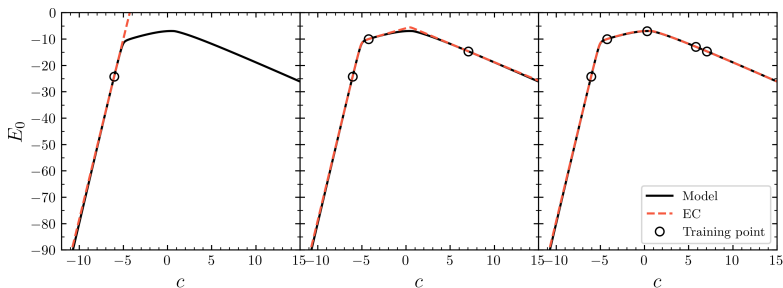


Eigenvector Continuation

- Projects the Hamiltonian to a subspace where its ground state can be rapidly evaluated
- For two training points c_1 and c_2 , with eigenvectors $|\psi_1\rangle$ and $|\psi_2\rangle$

$$M = \begin{bmatrix} \langle \psi_1 | H^0 | \psi_1 \rangle & \langle \psi_1 | H^0 | \psi_2 \rangle \\ \langle \psi_2 | H^0 | \psi_1 \rangle & \langle \psi_2 | H^0 | \psi_2 \rangle \end{bmatrix} + c \begin{bmatrix} \langle \psi_1 | H^1 | \psi_1 \rangle & \langle \psi_1 | H^1 | \psi_2 \rangle \\ \langle \psi_2 | H^1 | \psi_1 \rangle & \langle \psi_2 | H^1 | \psi_2 \rangle \end{bmatrix}$$

- Solve for smallest eigenvalue in the projected subspace for all values of the control parameter



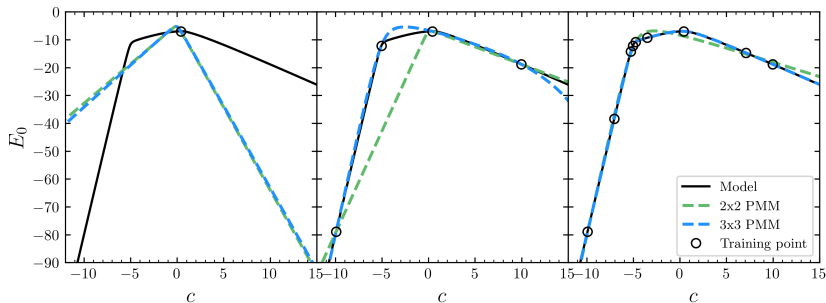
- Exact eigenstate requirement very tricky for AFDMC

Parametric Matrix Model

- Instead of explicit overlaps, fit matrix elements by minimizing cost function
- Dimensionality of PMM matrices is a free hyperparameter
- Can train against any number of training points

$$\tilde{H} = \begin{bmatrix} x_1 & 0 \\ 0 & x_2 \end{bmatrix} + c \begin{bmatrix} x_3 & x_4 \\ x_4 & x_5 \end{bmatrix}$$

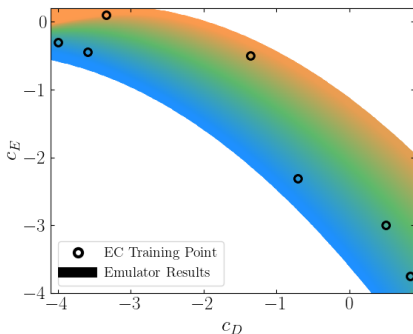
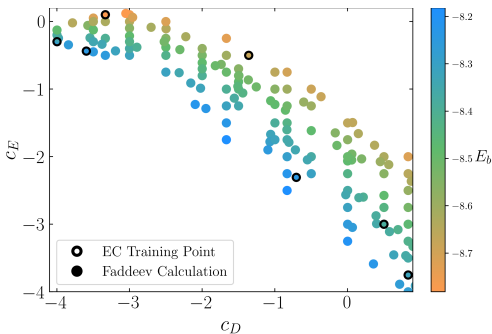
- Does not require any knowledge about exact eigenstates of the high-fidelity model (i.e. can be straightforwardly used for QMC)



Emulating Solutions to the Faddeev Equations

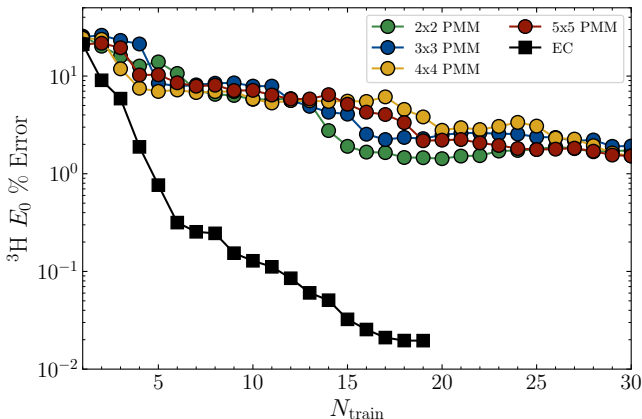
$$H = T + V_{NN} + V_{3N}^{TPE} + V_{3N}^D + V_{3N}^E$$
$$\tilde{H} = H^0 + C_S H^S + C_T H^T + \sum_{i=1}^7 C_i H^i + c_D H^D + c_E H^E$$

- High-cutoff interaction $R_0 = 0.6$ fm or $\Lambda_c \approx 660$ MeV
- Faddeev equations give exact solution for three-body wavefunction
- Solve for E_0 of ${}^3\text{H}$ for ~ 100 LEC sets (c_D/c_E shown for example)



Parametric Matrix Model vs Eigenvector Continuation

- Study how % error decreases with number of training points
- Additional computational overhead for eigenvector continuation (*seemingly* worth the effort)

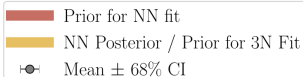
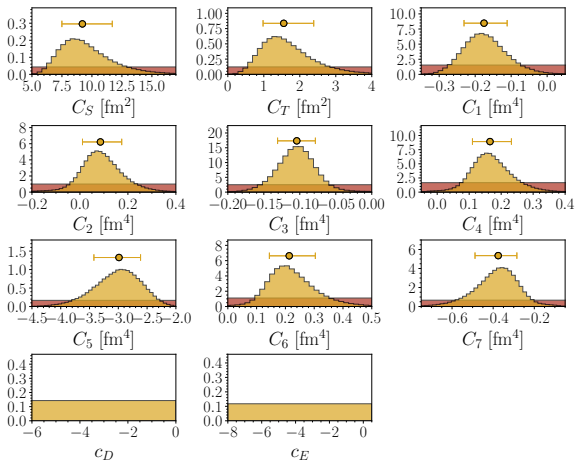


$$\mathcal{L} \propto \prod_i \exp \left[-\frac{1}{2} \left(\frac{X_i^{\text{exp.}} - X_i^{\text{theo.}}}{\sigma_i} \right)^2 \right]$$

- X_i includes ^3H binding energy, GT matrix element (Faddeev) and ^4He charge radius (AFDMC)
- $\sigma_i^2 = \sigma_{i,\text{exp}}^2 + \sigma_{i,\text{theo}}^2$
- Theoretical uncertainty $\sigma_{i,\text{theo}} = \Delta X_i$, estimated by performing calculations at lower chiral orders

$$\Delta X_{\text{N}^2\text{LO}}^{\text{EKM}} = \max \left[Q^4 |X_{\text{LO}}|, \quad Q^2 |X_{\text{LO}} - X_{\text{NLO}}|, \quad Q |X_{\text{NLO}} - X_{\text{N}^2\text{LO}}| \right]$$

Determining full N²LO LEC Posteriors

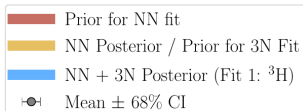
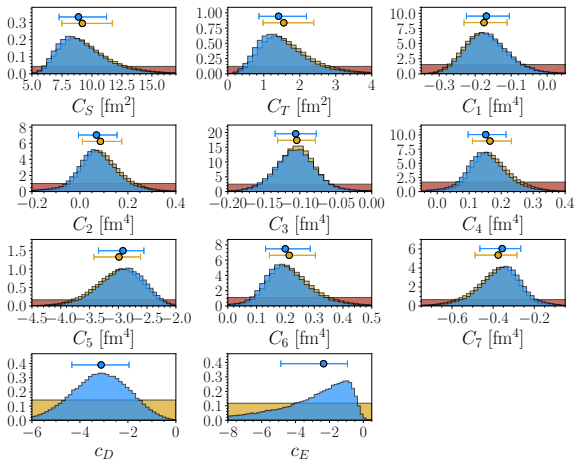


- Somasundaram *et. al* performed NN Bayesian fit at N²LO
- We take their posteriors as our priors for the NN LECs
- Include additional uniform priors on c_D and c_E

R. Somasundaram, J.E. Lynn, L. Huth, A. Schwenk, and I. Tews, Phys. Rev. C. **109**, 034005 (2024)

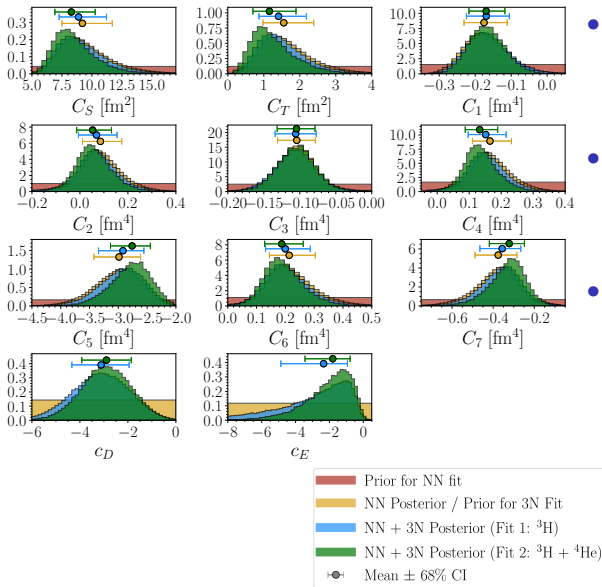
R. Curry, K. Hebeler, S. Gandolfi, A. Gezerlis, A. Schwenk, R. Somasundaram, and I. Tews
arXiv:2510.15860

Bayesian Fit Against ^3H



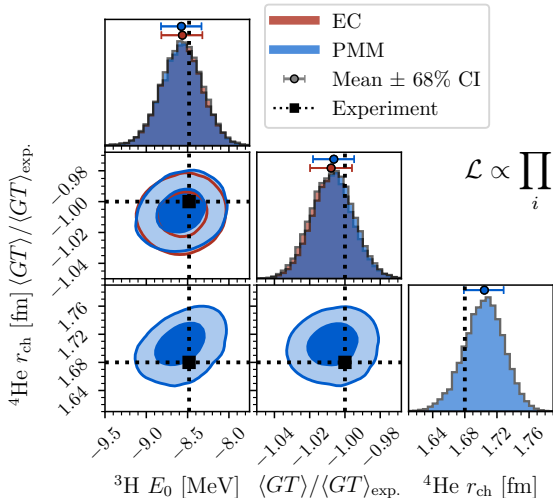
- N²LO posteriors when fit only against triton observables
- Likelihood factorizes, making this a global fit
- Very slight changes in NN LECs (strong dependence on prior)
- Long tail on c_E distribution

Additional ^4He Constraint



- AFDMC ^4He charge radius added to reduce long tail in c_E
- All NN LECs are also narrower due to its inclusion
- With LEC posteriors in hand, we can use them for many-body calculations

Posterior Predictions: PMM vs EC

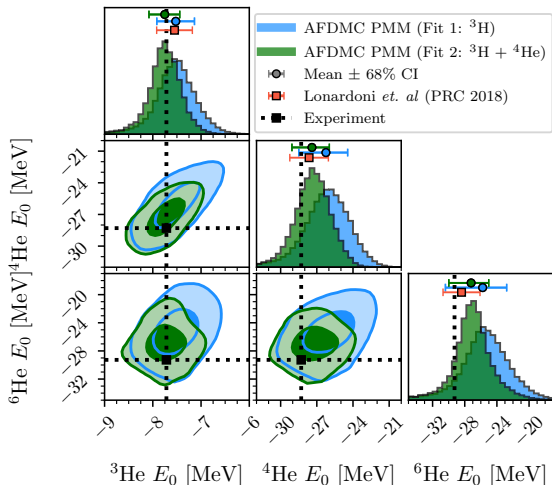


$$\mathcal{L} \propto \prod_i \exp \left[-\frac{1}{2} \left(\frac{X_i^{\text{exp.}} - X_i^{\text{theo.}}}{\sigma_i} \right)^2 \right]$$

- PMM $\equiv$ minimizing cost function
- EC $\equiv$ explicit overlap calculations

- PMM/EC give essentially identical results when used in likelihood

AFDMC calculations of helium isotopes



- Evaluate AFDMC emulators over 2.5×10^5 LEC samples
- Full posterior predictions for helium isotope binding energies
- Uncertainties propagated to many-body predictions
- Able to study correlations between different nuclei with QMC

Summary & Future Directions

- First Bayesian fit of local high-cutoff chiral EFT interactions containing all two- and three-nucleon operators at $N^2\text{LO}$
- Emulators employed at every step of the calculation (fitting the interaction $\rightarrow$ final many-body predictions)
- Opportunity to explore other fitting observables
- Framework for future Quantum Monte Carlo studies of more light- and medium-mass nuclei with fully propagated theoretical uncertainties

Thank you for listening!

Collaborators:

- Alex Gezerlis (Guelph)
- Ingo Tews (LANL)
- Stefano Gandolfi (LANL)
- Rahul Somasundaram (LANL)
- Achim Schwenk (TUD)
- Kai Hebeler (TUD)

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